

LECTURE

WHAT IS A BOX PLOT?

SECTION: CREATE A BOX PLOT ON CHATGPT & JULIUS AI



IN THIS LECTURE, YOU WILL LEARN

WHAT IS A BOX PLOT?

WHY IS IT IMPORTANT?

HOW IS THE BOX PLOT USED IN SIX SIGMA PROJECTS?

WHAT ARE THE COMPONENTS OF A BOX PLOT?

UNDERSTANDING THE VARIABLES IN A BOX PLOT

CHAT GPT

WHAT IS A BOX PLOT?

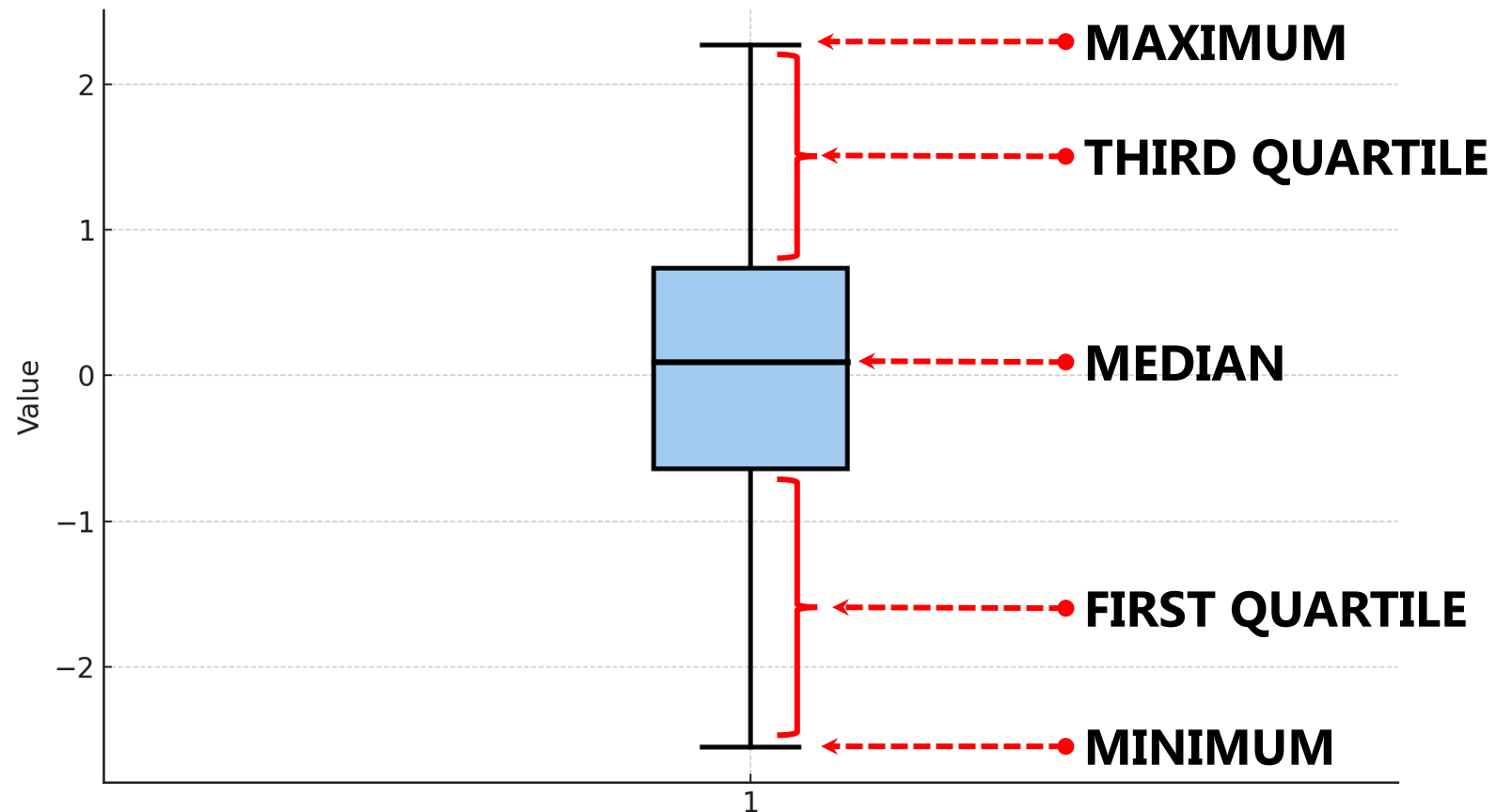
WHAT IS A BOX PLOT?

ALSO KNOWN AS THE BOX AND WHISKER PLOT

DISPLAYS THE DISTRIBUTION OF DATA BASED ON THE FOLLOWING:

REVEALS:

- **CENTRAL TENDENCY**
- **DISPERSION**
- **SKEWNESS OF DATA**
- **HIGHLIGHT OUTLIERS**



CHAT GPT

WHY IS IT IMPORTANT?

WHY IS A BOX PLOT IMPORTANT?

CONCISE DATA SUMMARY

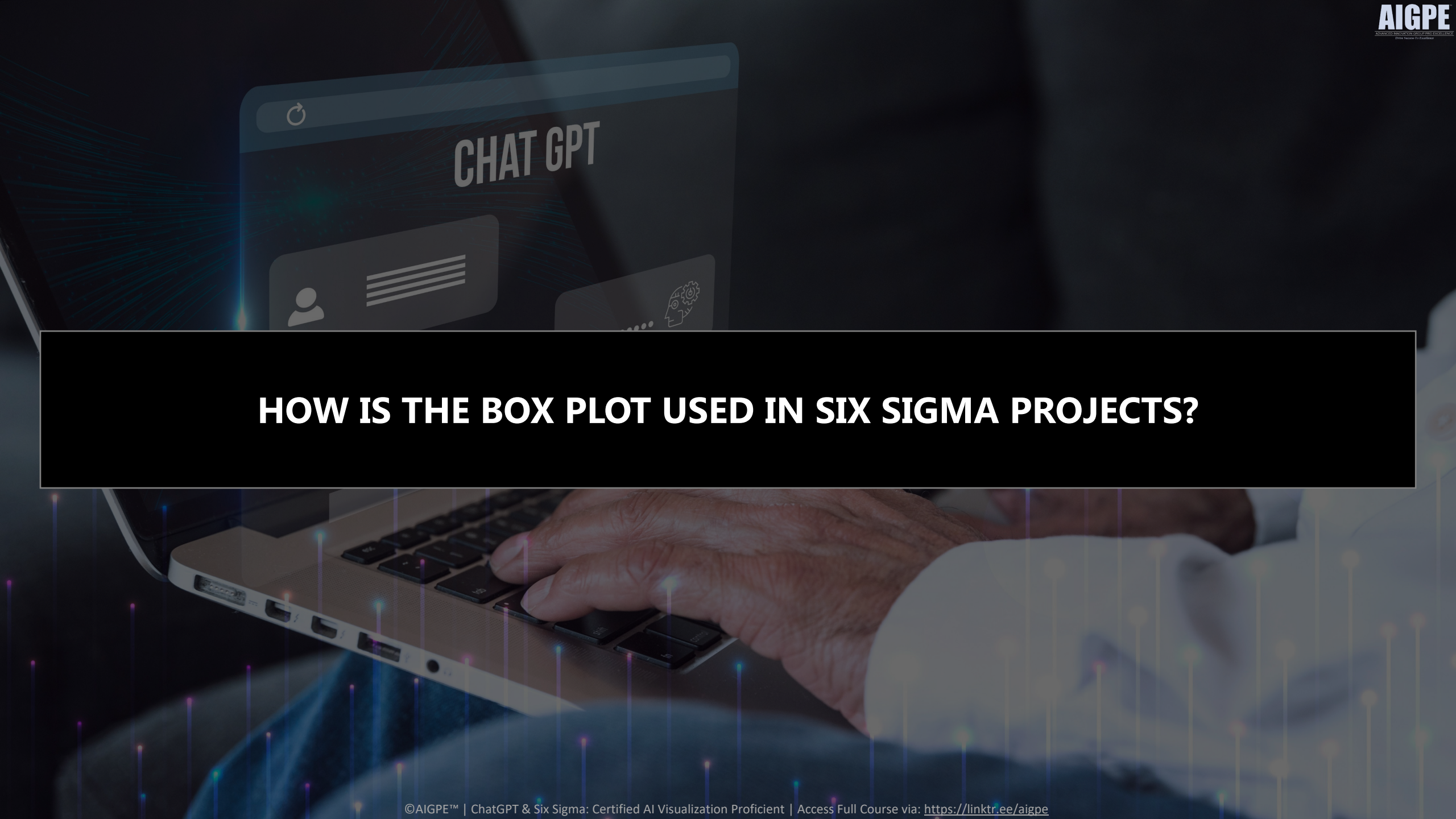
- **HIGH-LEVEL EXECUTIVE SUMMARY OF YOUR DATA**
- **PROVIDES AVERAGE, RANGE & OVERALL SPREAD**
- **ALLOWS SWIFT, INFORMED DECISIONS**

COMPARISON ACROSS GROUPS

- **PROVIDES VISUAL PERFORMANCE REPORTS SIDE-BY-SIDE**
- **SHOWS HOW EACH GROUP IS DOING COMPARED TO OTHERS**
- **HELPS IDENTIFY AREAS THAT EXCEL & THOSE THAT NEED MORE ATTENTION**

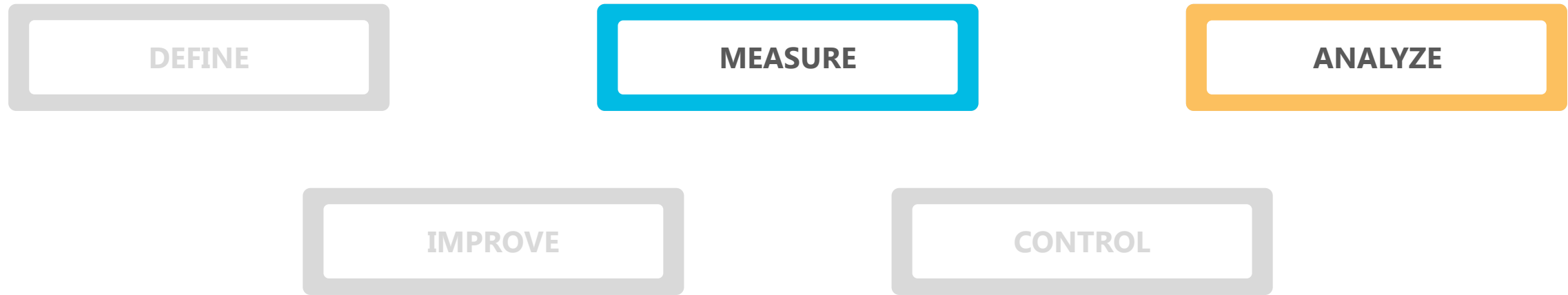
OUTLIER IDENTIFICATION

- **UNEXPECTED EXPENSES OR SUDDEN PROFITS IN BUSINESS**
- **UNUSUAL POINTS THAT DON'T FOLLOW USUAL PATTERN**
- **HELPS SPOT UNUSUAL OR EXTREME VALUES**

The background image shows a person's hand typing on a laptop keyboard. Overlaid on the laptop screen is a semi-transparent interface for 'CHAT GPT'. The interface includes a refresh icon in the top left, the text 'CHAT GPT' in a large, bold, sans-serif font, and two chat bubbles below it. The first bubble contains a person icon and three horizontal lines, while the second bubble contains a gear icon and a head profile with gears inside. The entire scene is set against a dark, blurred background with some digital light effects.

HOW IS THE BOX PLOT USED IN SIX SIGMA PROJECTS?

HOW IS THE BOX PLOT USED IN SIX SIGMA PROJECTS?



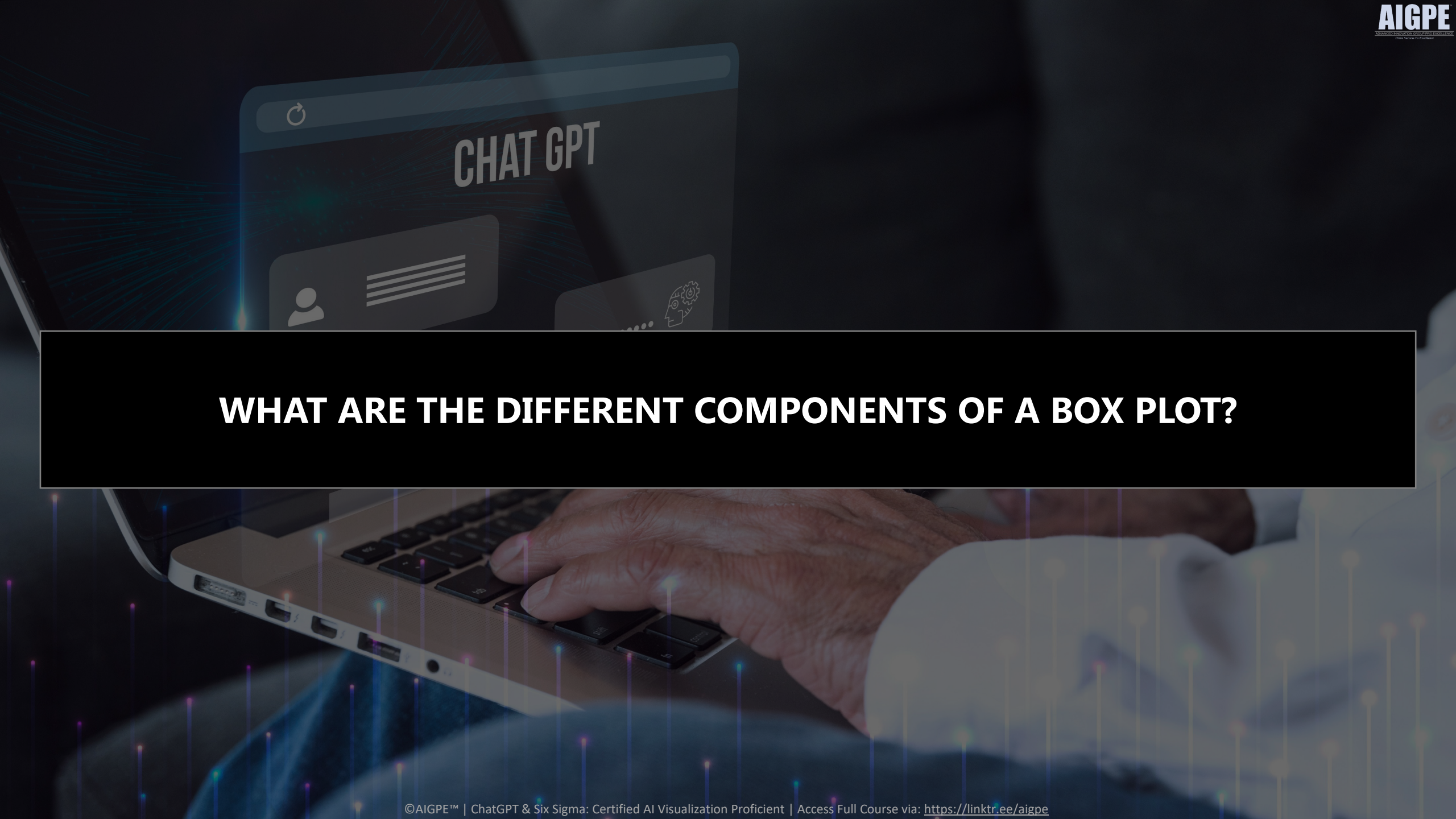
VISUALLY ASSESS THE DISTRIBUTION OF DATA

PROVIDES AN UNDERSTANDING OF DATA'S SPREAD, CENTRAL TENDENCY, AND OUTLIERS

HELPS DETERMINE AREAS THAT NEED IMPROVEMENT

EFFECTIVELY IDENTIFIES OUTLIERS & RANGE OF VARIABILITY IN PROCESS PERFORMANCE

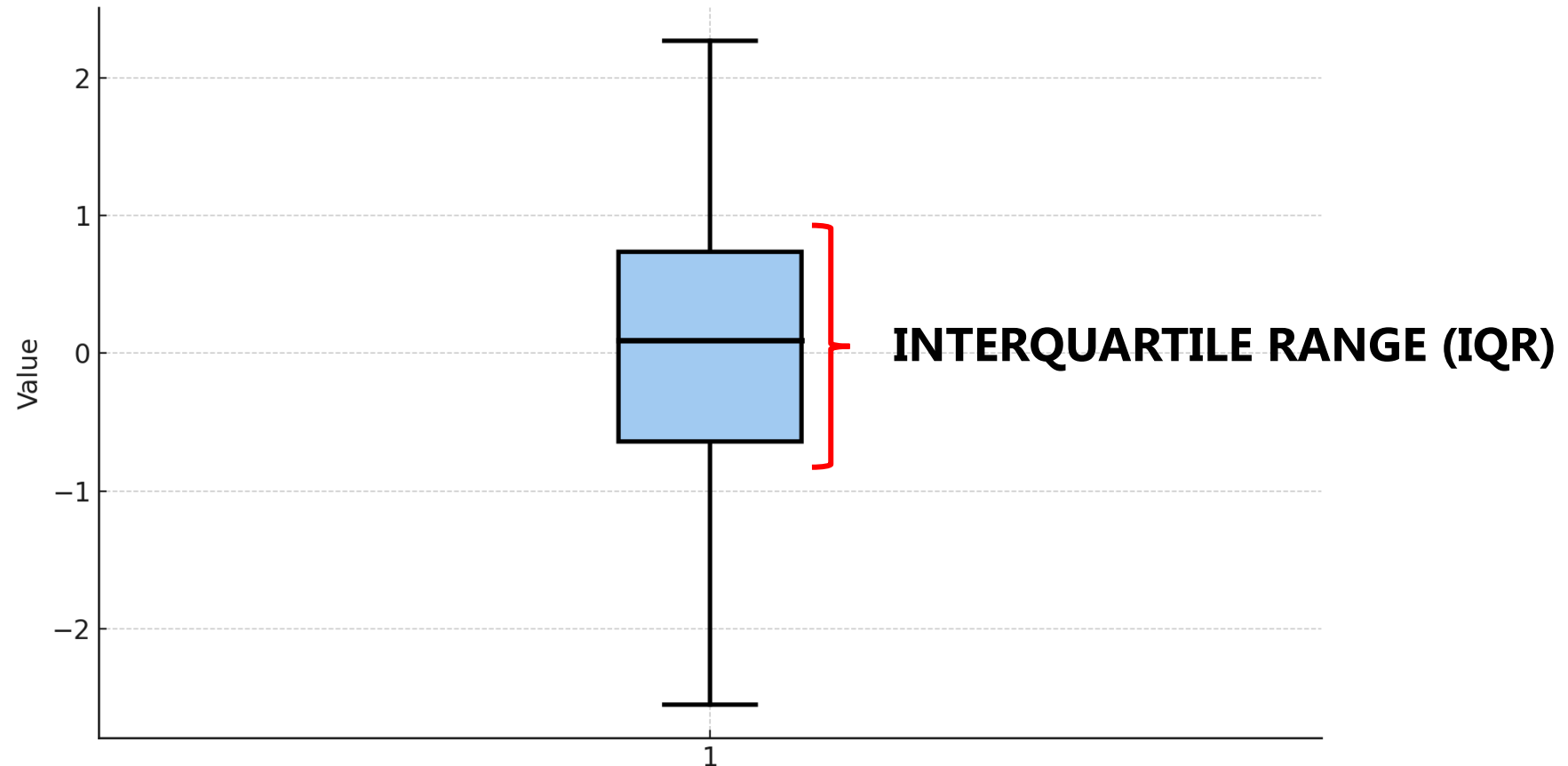
PINPOINTS SPECIFIC AREAS WHERE EFFORTS SHOULD BE CONCENTRATED

The background image shows a person's hand typing on a laptop keyboard. Overlaid on the laptop screen is a semi-transparent interface for 'CHAT GPT'. The interface includes a refresh icon in the top left, the text 'CHAT GPT' in large letters, and two chat bubbles below it. The first bubble contains a person icon and three horizontal lines, while the second bubble contains a gear icon and a head profile with gears inside. The entire scene is set against a dark background with faint, colorful light trails.

WHAT ARE THE DIFFERENT COMPONENTS OF A BOX PLOT?

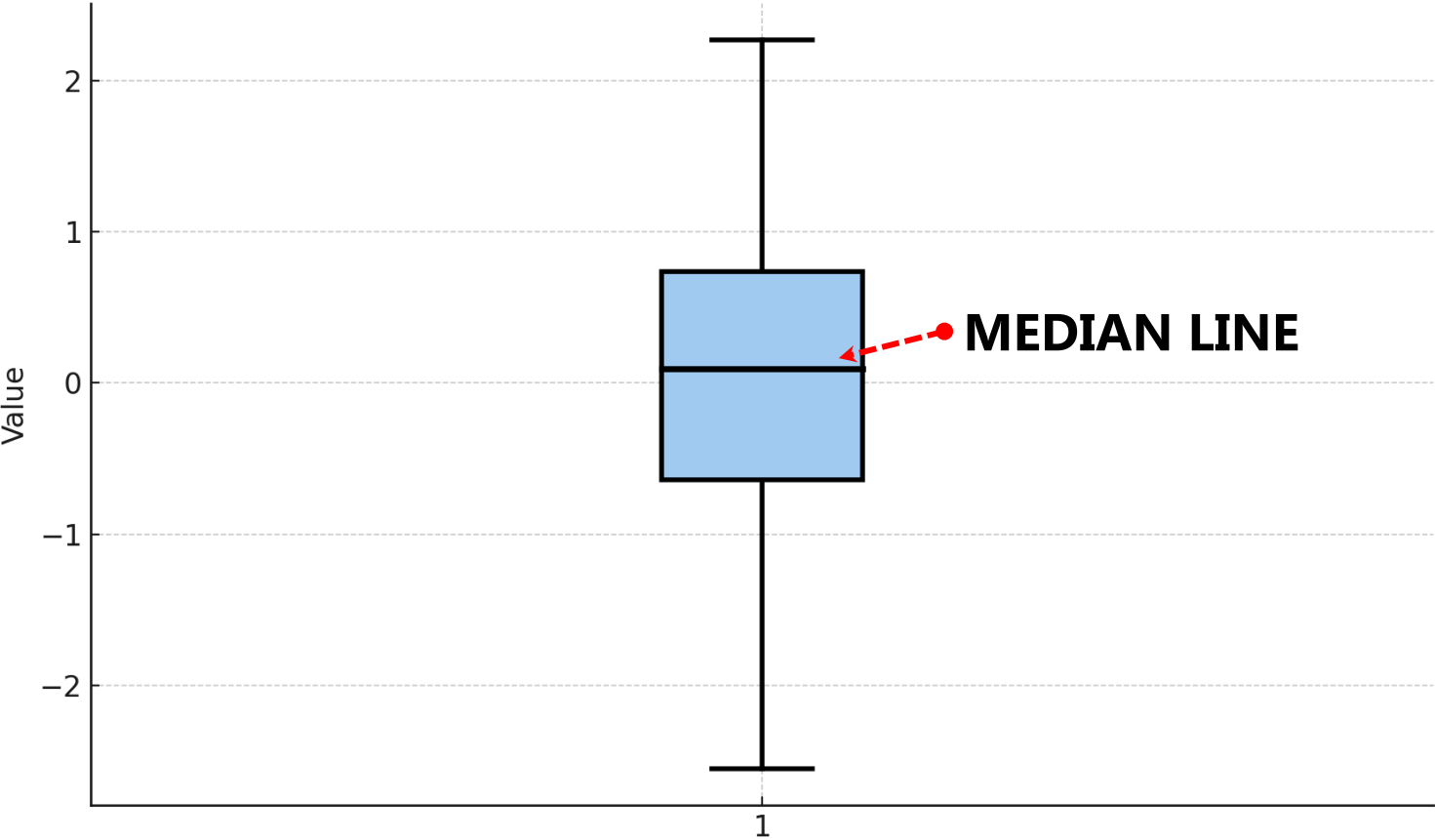
REPRESENTS INTERQUARTILE RANGE (IQR), THE MIDDLE 50% OF THE DATASET

BOTTOM AND TOP EDGES OF THE BOX INDICATE THE FIRST & THIRD QUANTILES



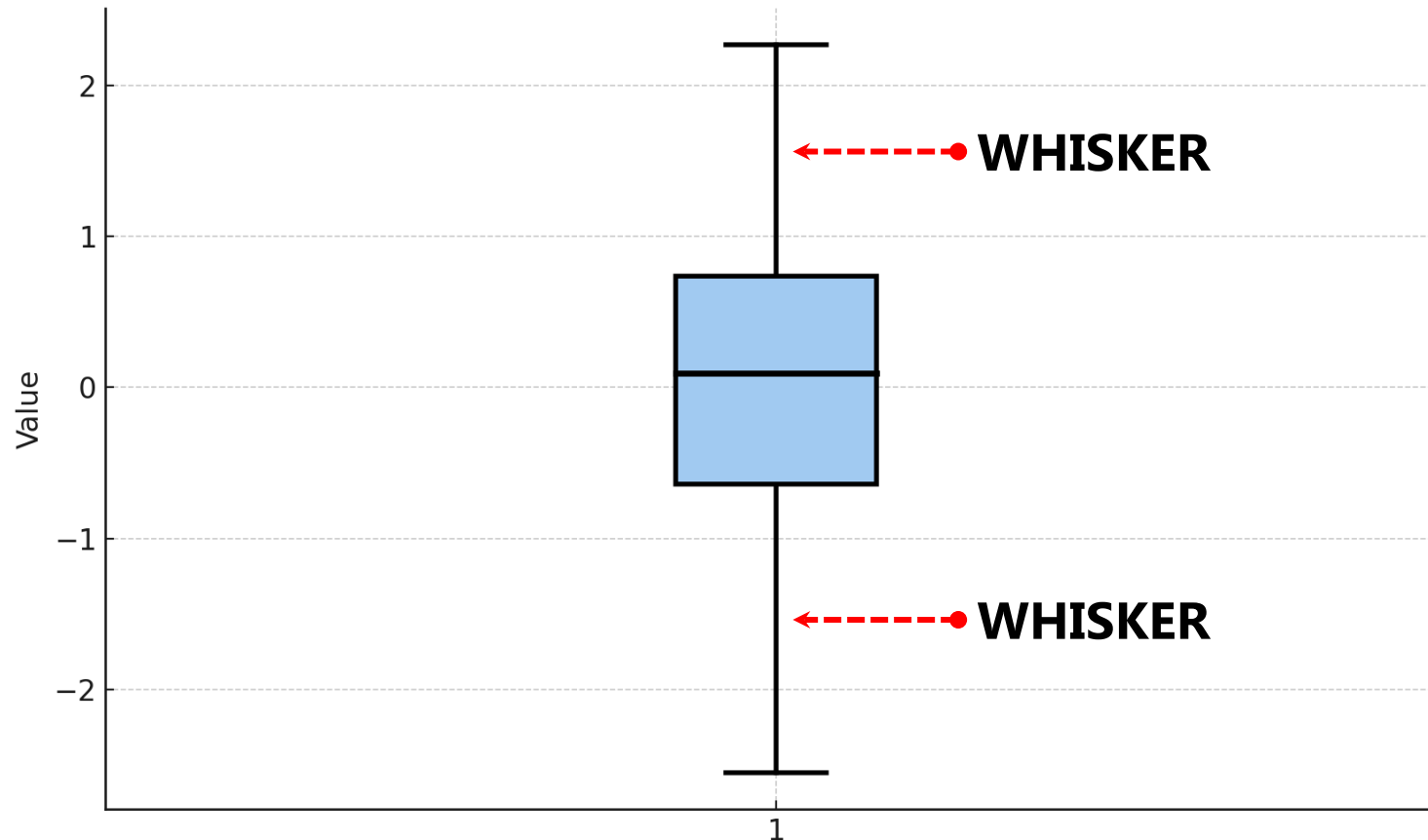
THE MEDIAN LINE

THE LINE ACROSS THE BOX INDICATES THE DATASET’S MEDIAN



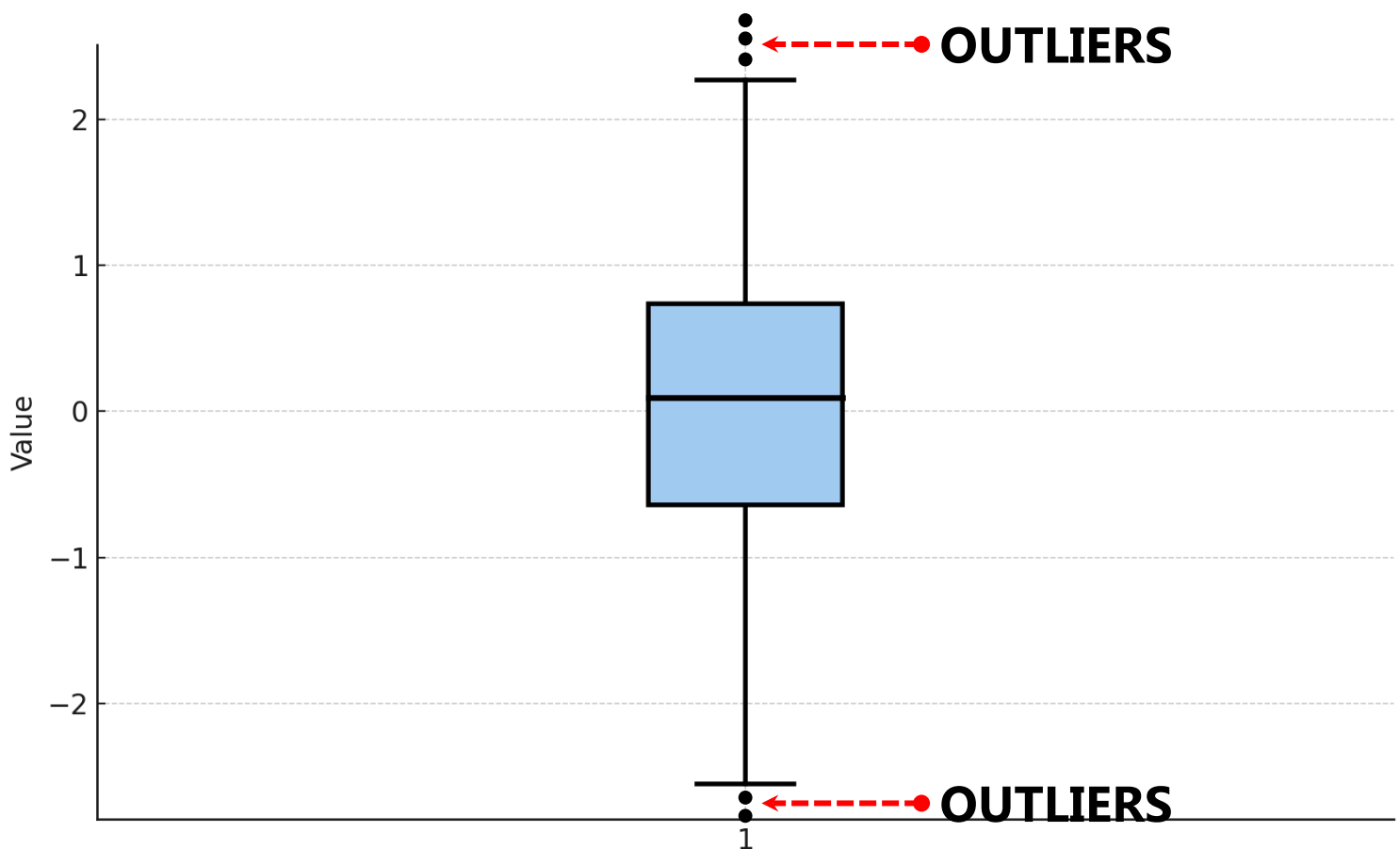
Lines extending from the box represent the range outside the first & third quartiles

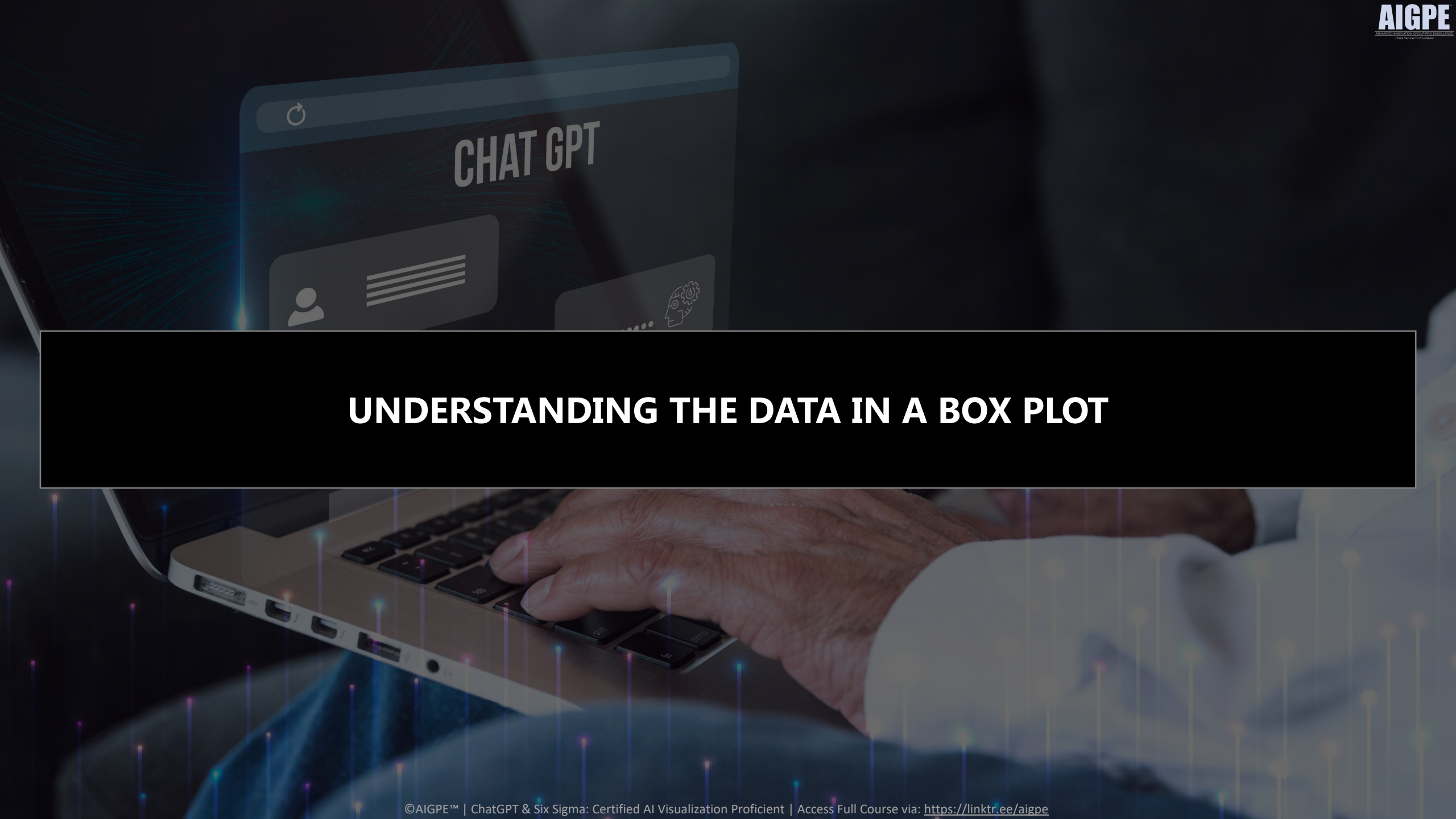
Typically extend '1.5 X IQR' above & below the box



OUTLIERS

POINTS PLOTTED BEYOND THE WHISKERS ARE TERMED AS OUTLIERS



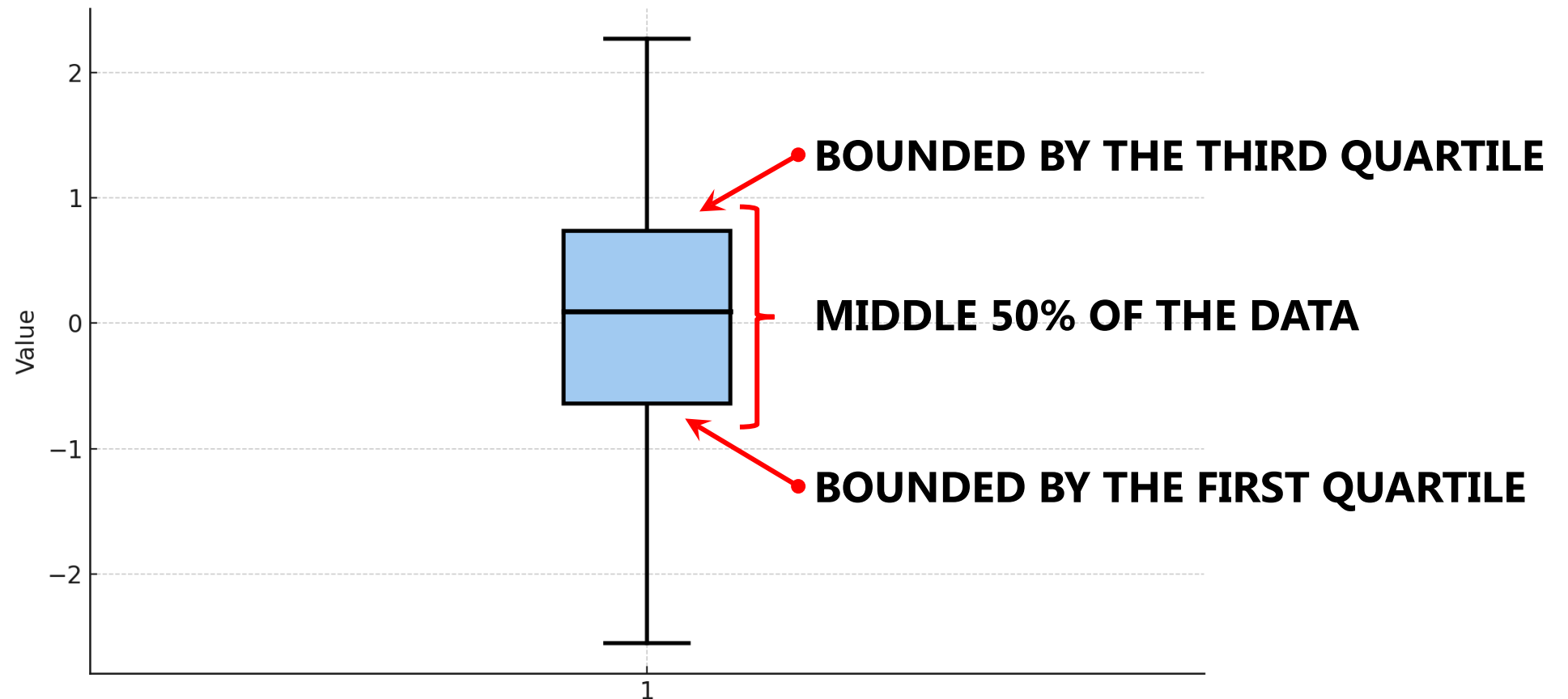
The background image shows a person's hand typing on a laptop keyboard. Overlaid on the laptop screen is a semi-transparent interface for 'CHAT GPT'. The interface includes a refresh icon in the top left, the text 'CHAT GPT' in large letters, and two chat bubbles below it. The first bubble contains a person icon and three horizontal lines, while the second bubble contains a gear icon and a head profile with gears. The entire scene is set against a dark background with abstract blue and purple light streaks and particle effects.

UNDERSTANDING THE DATA IN A BOX PLOT

UNDERSTANDING THE DATA IN A BOX PLOT

THE CENTRAL BOX REPRESENTS THE MIDDLE 50% OF THE DATA

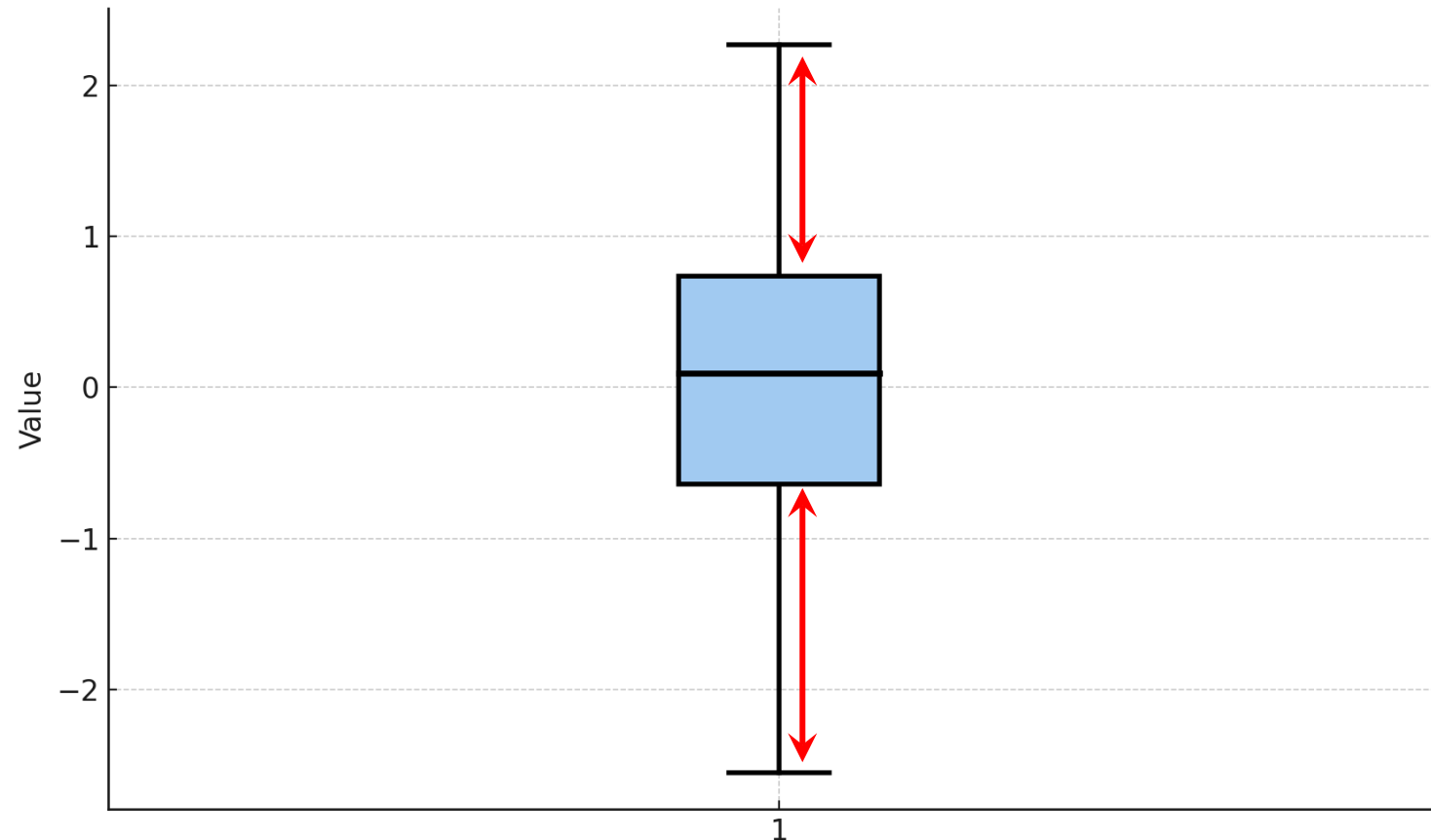
IT IS BOUNDED BY THE FIRST AND THIRD QUANTILES



UNDERSTANDING THE DATA IN A BOX PLOT

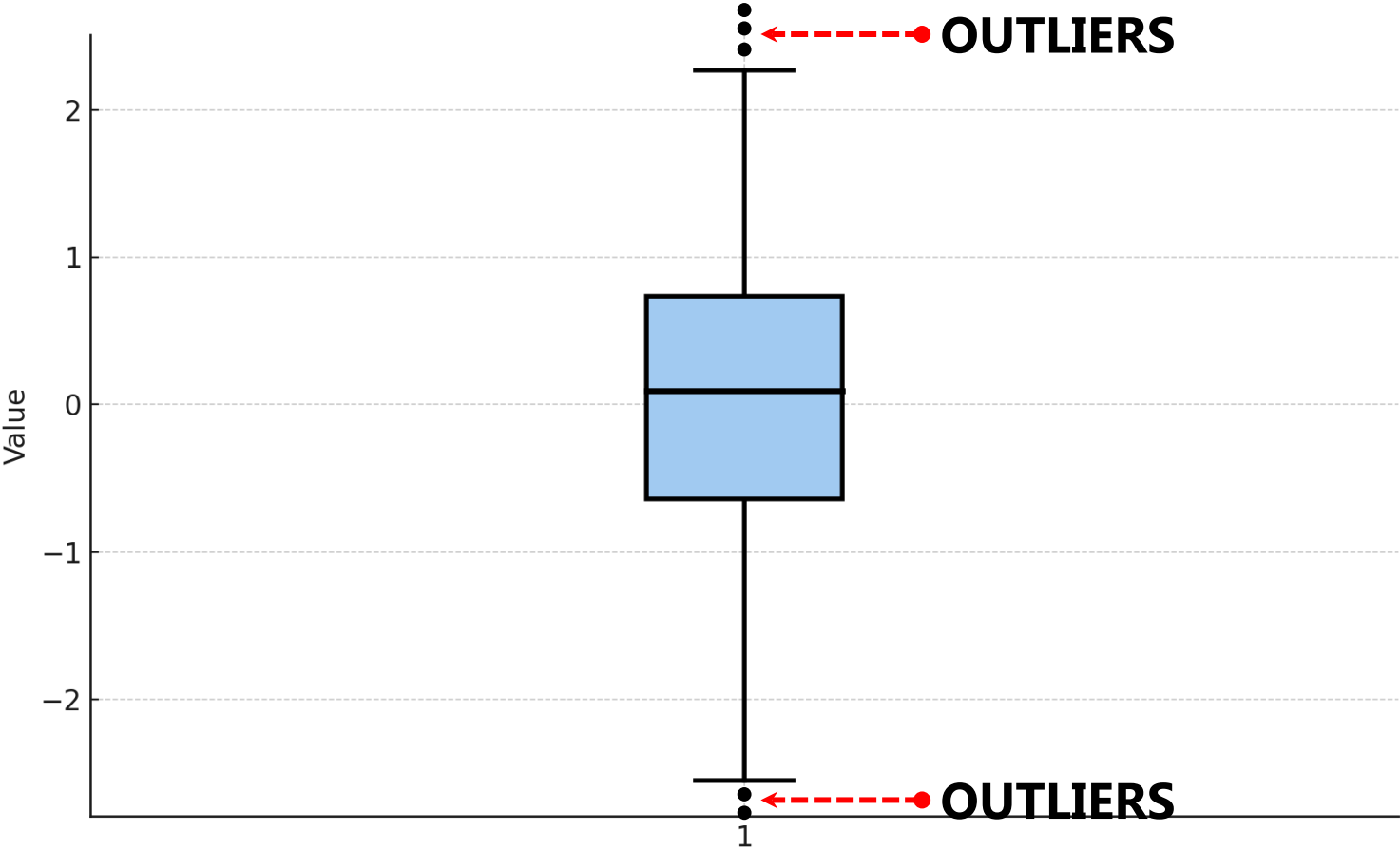
WHISKERS EXTEND TO THE MOST EXTREME DATA POINTS

PROVIDES A SENSE OF THE DATA'S SPREAD



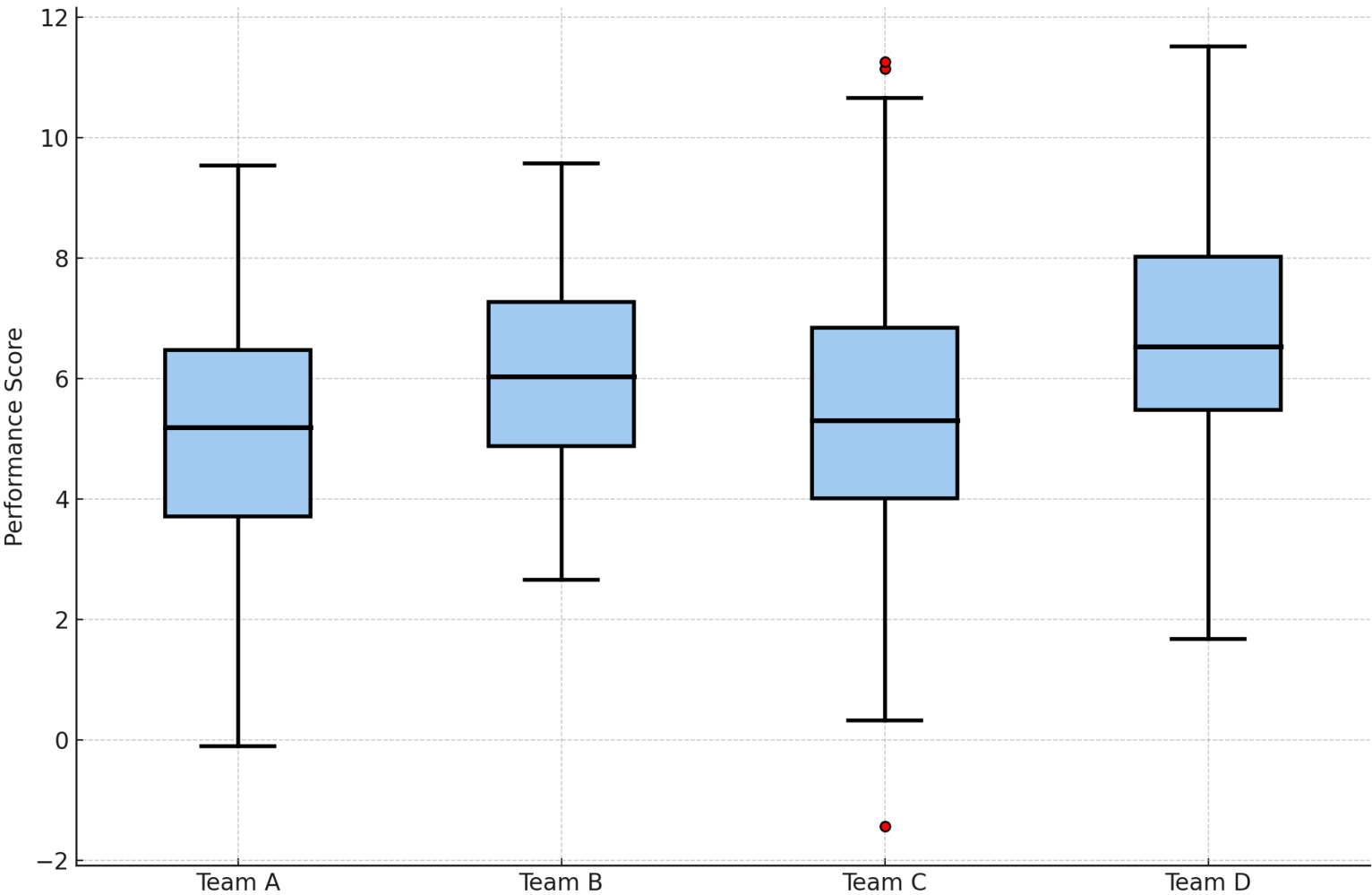
UNDERSTANDING THE DATA IN A BOX PLOT

OUTLIERS ARE DISPLAYED AS INDIVIDUAL DATA POINTS BEYOND THE WHISKERS



UNDERSTANDING THE DATA IN A BOX PLOT

EASILY COMPARES DIFFERENT DATA SETS & QUICKLY IDENTIFIES MEDIAN, RANGE & OUTLIERS



AIGPE™

ADVANCED INNOVATION GROUP PRO EXCELLENCE

Drive Success To Excellence